

- (ii) C marked at (0.032, 0.2) **B1**
- 6 (a) (i) cannot predict when a particular nucleus will decay
or **B1**
cannot predict which nucleus will decay next
- (ii) (decay is) not affected by external (environmental) factors **B1**
- (b) fluctuations in (measured) count rate **B1**
- (c) The average time taken for the initial number of nuclei (or activity) of that particular radioactive nuclide to reduce to half of its initial value. **B1**
- (d) (i) activity (of X at time t) **B1**

(ii)

B3

- Y is a stable isotope
- total number of nuclei is constant
- half-life (of X) is 13.6 s
- decay constant (of X) is 0.051 s^{-1} $\left[\ln 2 / t_{\frac{1}{2}} = \ln 2 / 13.6 \text{ s} = 0.051 \text{ s}^{-1} \right]$
- amount (of X) at $t = 0$ is 0.066 mol
 $\left[n_0 = N_0 / N_A = 4.00 \times 10^{22} / 6.02 \times 10^{23} = 0.066 \text{ mole} \right]$
- activity (of X) at $t = 0$ is $2.0 \times 10^{21} \text{ Bq}$
 $\left[A_0 = \lambda N_0 = (0.051)(4.00 \times 10^{22}) = 2.0 \times 10^{21} \text{ Bq} \right]$

Any three points, 1 mark each

(e) mass of 1 nucleus = $(7.3 \times 10^{-4}) / (4.0 \times 10^{22})$

C1

nucleon number = mass of nucleus / (1.66×10^{-27})

C1

= $(7.3 \times 10^{-4}) / (4.0 \times 10^{22} \times 1.66 \times 10^{-27})$

= 11 **and** given as an integer

A1